# TS2000
# TS2100
Robot Controller

# INSTRUCTION MANUAL

## TS2000 ROBOT CONTROLLER

## STARTUP MANUAL

Rev.5
April 1, 2020

**Notice**

- Make sure that this instruction manual is delivered to the final user of SHIBAURA MACHINE's industrial robot.
- Before operating the industrial robot, read through and completely understand this manual.
- After reading through this manual, keep it nearby for future reference.

Shibaura Machine

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# Preface

This manual describes the procedures covering from power ON of the robot after installation to robot operation by manual guide and by programs.
Before reading this manual, install the robot according to the Safety Manual and Installation & Transport Manual.
When you use the TS2000 robot controller for the first time, be sure to read through this manual.

Structure of TS2000 robot controller instruction manuals:
These instruction manuals were published in parts according to the application and purpose, and the name and outline of each manual are as follows:

[Safety Manual]

This manual contains the important information to use the robot safety and correctly. Be sure to read through and understand this manual before operating the robot. Also, strictly observe the descriptions made there.

[Installation & Transport Manual]

This manual describes the transport, unpacking and installation of the robot and controller. Be sure to read through this manual before unpacking the shipment containing the robot.

[Operator's Manual]

This manual deals with the TS1000 controller operating procedures. Read through this manual before operating the robot, and refer to it as necessary.

[Robot Language Manual]

This manual refers to the robot language called "SCOL". When you have to create a program based on this language, read through the manual.

[Interface Manual]

This manual describes the external signals for the robot. Concerning the interface conditions between the robot and peripheral equipment, specifications, timing, etc., refer to the manual when necessary.

[Maintenance Manual]

This manual deals with the daily and regular inspections to be made on the robot and controller. Read through this manual to use the robot safely over long years to come.

[Communication Manual]

This manual describes the serial communication between the robot controller and other equipment. Refer to this manual when connecting the robot controller with a host computer, optical sensor, etc., via a serial cable.

# Cautions on Safety

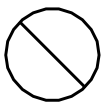
This manual contains the important information on the robot and controller to prevent injury to the operators and persons nearby, to prevent damage to assets and to assure correct use.
Make sure that the following details (indications and symbols) are well understood before reading this manual. Always observe the information that is noted.

[Explanation of indications]

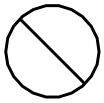
| Indication | Meaning of indication |
|---|---|
| ! **DANGER** | This means that "incorrect handling will lead to fatalities or serious injuries." |
| ! **CAUTION** | This means that "incorrect handling may lead to personal injuries *1) or physical damage *2). |

*1) Injuries refer to injuries, burns and electric shocks, etc., which do not require hospitalization or long-term medical treatment.
*2) Physical damage refers to damages due to destruction of assets or resources.

[Explanation of symbols]

| Symbol | Meaning of symbol |
|---|---|
|  | This means that the action is prohibited (must not be done). Details of the actions actually prohibited are indicated with pictures or words in or near the symbol. |
|  | This means that the action is mandatory (must be done). Details of the actions that must be done are indicated with pictures or words in or near the symbol. |
|  | This means danger. Details of the actual danger are indicated with pictures or words in or near the symbol. |
|  | This means caution. Details of the actual caution are indicated with pictures or words in or near the symbol. |

[Operation]

| ! DANGER | |
|---|---|
|  Prohibited | • During operation, NEVER enter the dangerous area of the robot. Otherwise, you will be injured seriously.<br>• DO NOT leave in the working range any machinery or materials which will hinder the operation. If the equipment went wrong, a person nearby will be injured or involved in an accident.<br>• Anyone other than the operator MUST NOT approach the equipment. Should he negligently touch the dangerous part of the equipment, he will get injured or involved in a serious accident.<br>• NEVER perform an inappropriate operation which is not described in the instruction manual. Otherwise, the equipment will start by mistake, resulting in personal injury or serious accident. |
|  Danger | • If you feel even a little that you are exposed to danger or that the equipment works abnormally, press the EMERGENCY stop pushbutton switch to stop the equipment. If the equipment is used as it is, you will be injured or involved in a serious accident. When this happens, ask our after-sale service agent for repair.<br>• During operation, be sure to close the equipment cover. Should the cover be opened during operation, you will be struck by an electric shock or get injured.<br>• Only a well-trained and qualified person is allowed to perform the operation. Should the equipment be operated improperly, it will start by mistake, causing a personal injury or serious accident.<br>• If the equipment has malfunctioned, turn the power off, identify and remove the cause of the abnormality, maintain the peripheral equipment and completely restore the malfunctioned equipment. Then start the equipment at a slow speed. If the equipment starts, leaving the abnormality, you will be involved in a serious accident. |

| ! CAUTION | |
|---|---|
| Prohibited | • DO NOT change the data of the system structure file. Otherwise, the robot will move abnormally, resulting in damage or an accident. |
| Mandatory | • In principle, teaching operation should be performed outside the dangerous area of the robot. If it should be performed inevitably within the dangerous area, strictly observe the following matters.<br>(1) The teaching operation should always be performed by two (2) persons. One person performs the job and the other person watches outside the dangerous area. Also, both persons should try to prevent mis-operation with each other.<br>(2) The operator should do the job in an attitude ready to press the EMERGENCY stop pushbutton switch at any time. Also, he should perform the job at a position from which he can evacuate immediately at the time of an emergency after confirming the robot working range and shields nearby.<br>(3) The supervisor should keep watch on the job at a position where he can see the entire robot system and operate the EMERGENCY stop pushbutton switch at the time of an emergency. Also, he should keep anyone from entering the dangerous area. Unless the operator or other person follows the instructions of the supervisor, an accident will be caused.<br>• If an abnormality has generated or the POWER LED lamp on the control panel remains off after the main power switch of the equipment was turned on, turn off the main power immediately and confirm the wiring. Otherwise, you will be struck by an electric shock or a fire will break out.<br>• Unless the robot operates toward a designated direction at manual guide, turn off the servo power. Otherwise, the robot will be damaged or you will be involved in an accident. When this happens, call us at the after-sale service agent.<br>• Pushbutton operations on the operation panel and teach pendant should be confirmed visually. Otherwise, you will be involved in an accident due to mis-operation.<br>• After the power is turned on, be sure to reset a program to start an automatic operation. If the program is executed continuously, the robot will interfere with the peripheral equipment, resulting in damages or accidents. |

| CAUTION | |
|---|---|
|  Mandatory | • Before operating the equipment, perform the following inspection.<br>(1) Make sure that visual appearance of the robot, controller, peripheral equipment and cables is in the good condition.<br>(2) Make sure that no obstacle stands in or near the working range of the robot and peripheral equipment.<br>(3) Make sure that the emergency stop and other safety devices operate properly.<br>(4) Make sure that no abnormal noise or vibration is involved in the robot operation.<br>If the above prior inspection is skipped, the equipment will be damaged or you will be involved in an accident. |
|  Caution | • The speed of test operations is initially set at 25 % of the maximum robot speed.<br>• The speed of automatic operation is initially set at 100 % of the maximum robot speed. |

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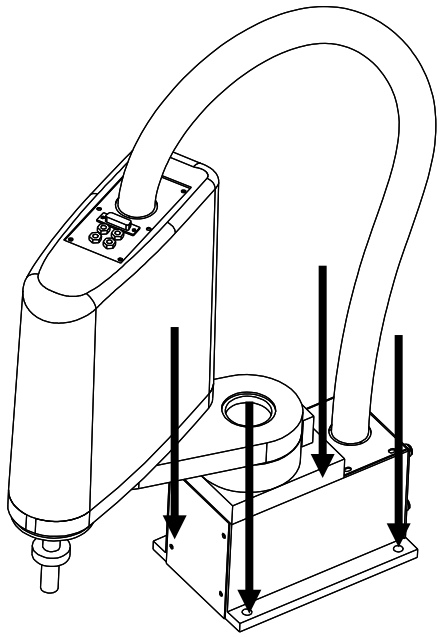
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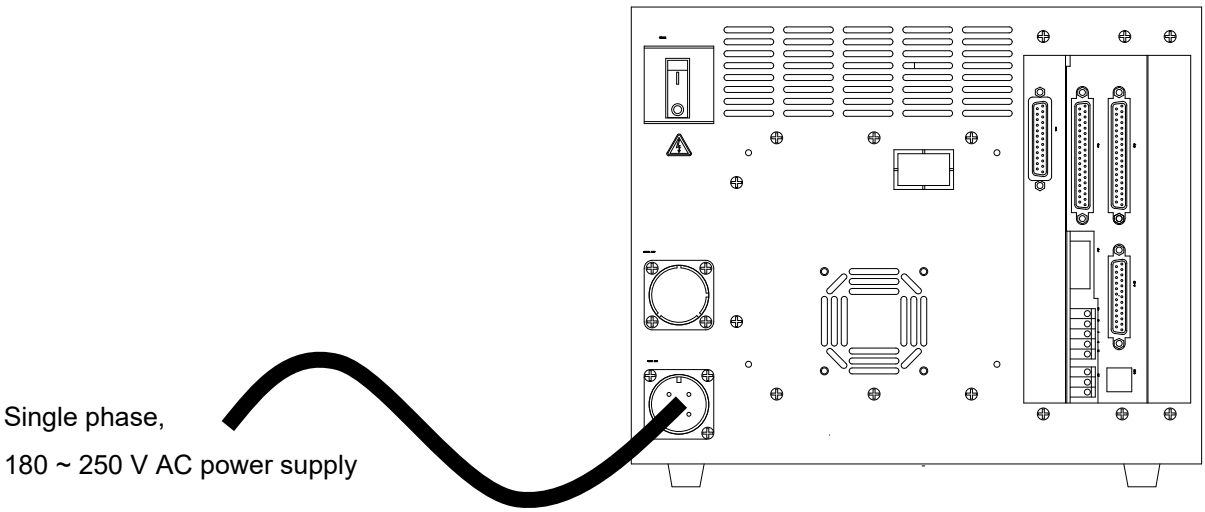
## 1. Before Power ON

Before turning the power on, confirm the following matters, referring to the Installation & Transport Manual.

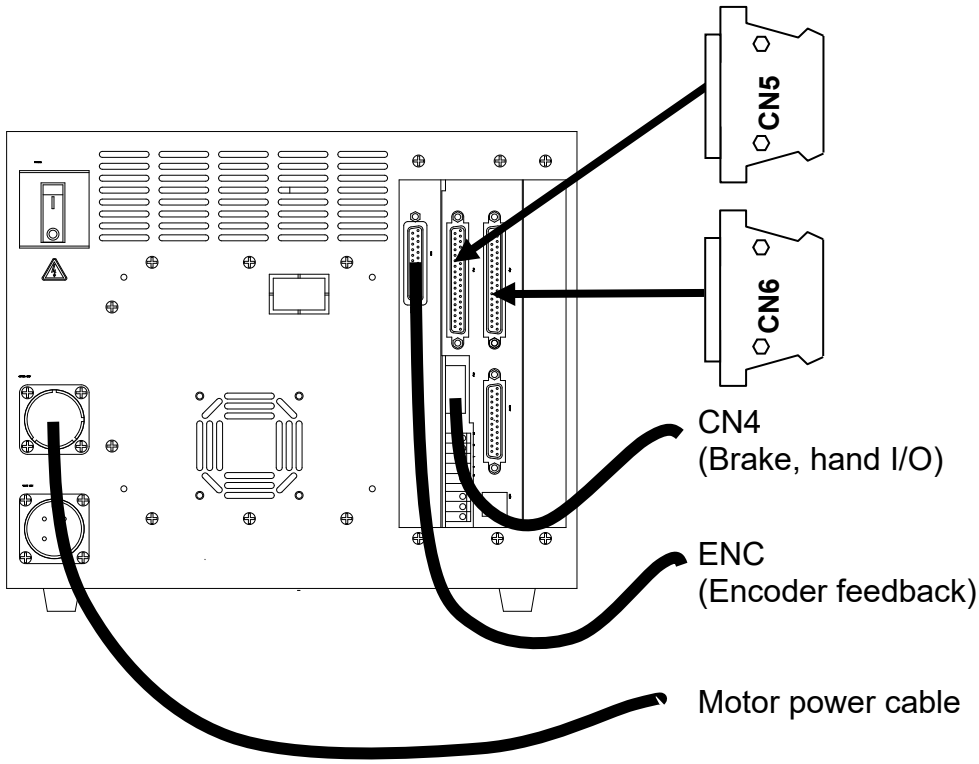
Is the robot secured?



Is the power supply connected?

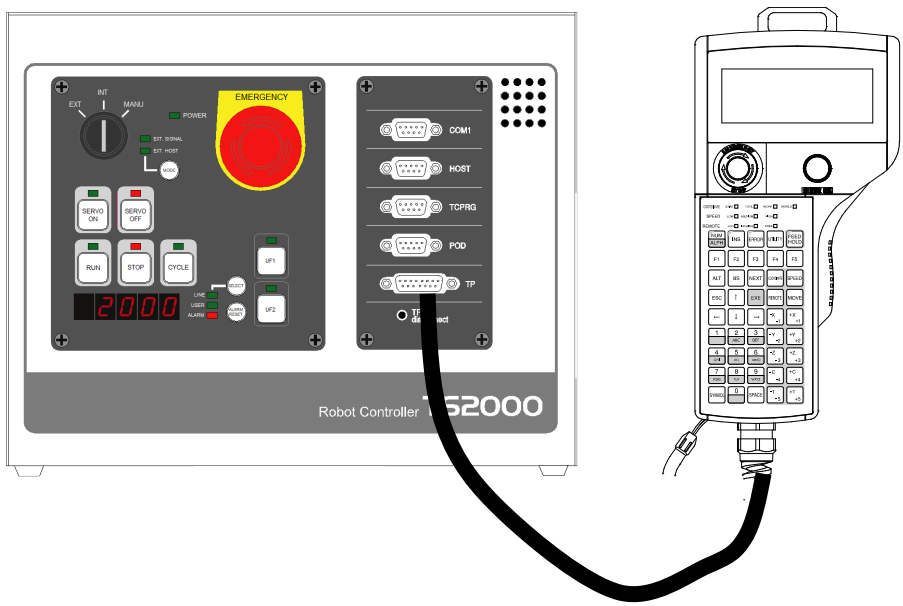


Are the cables between the controller and robot connected?
Is the I/P jumper connector connected?



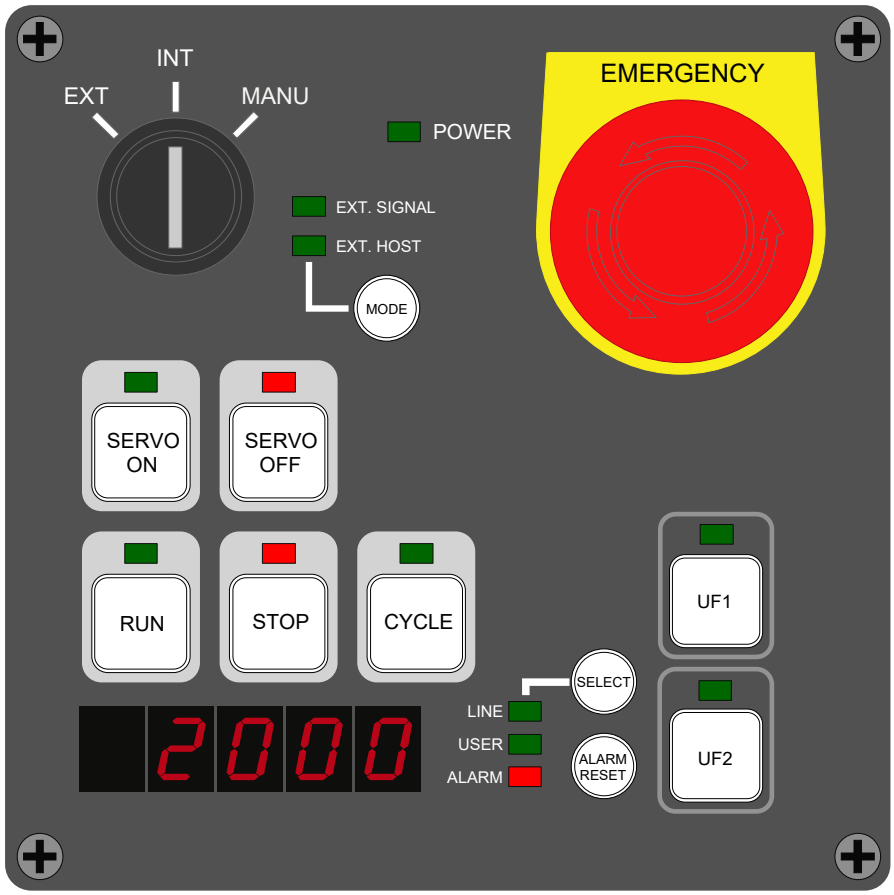

The I/O jumper connector is short-circuited to the external signals required for the robot operation, such as external emergency stop, servo OFF and stop. The robot can be moved only when these signals are connected.

Is the teach pendant (TP) connected?



## 1.1 Control Panel

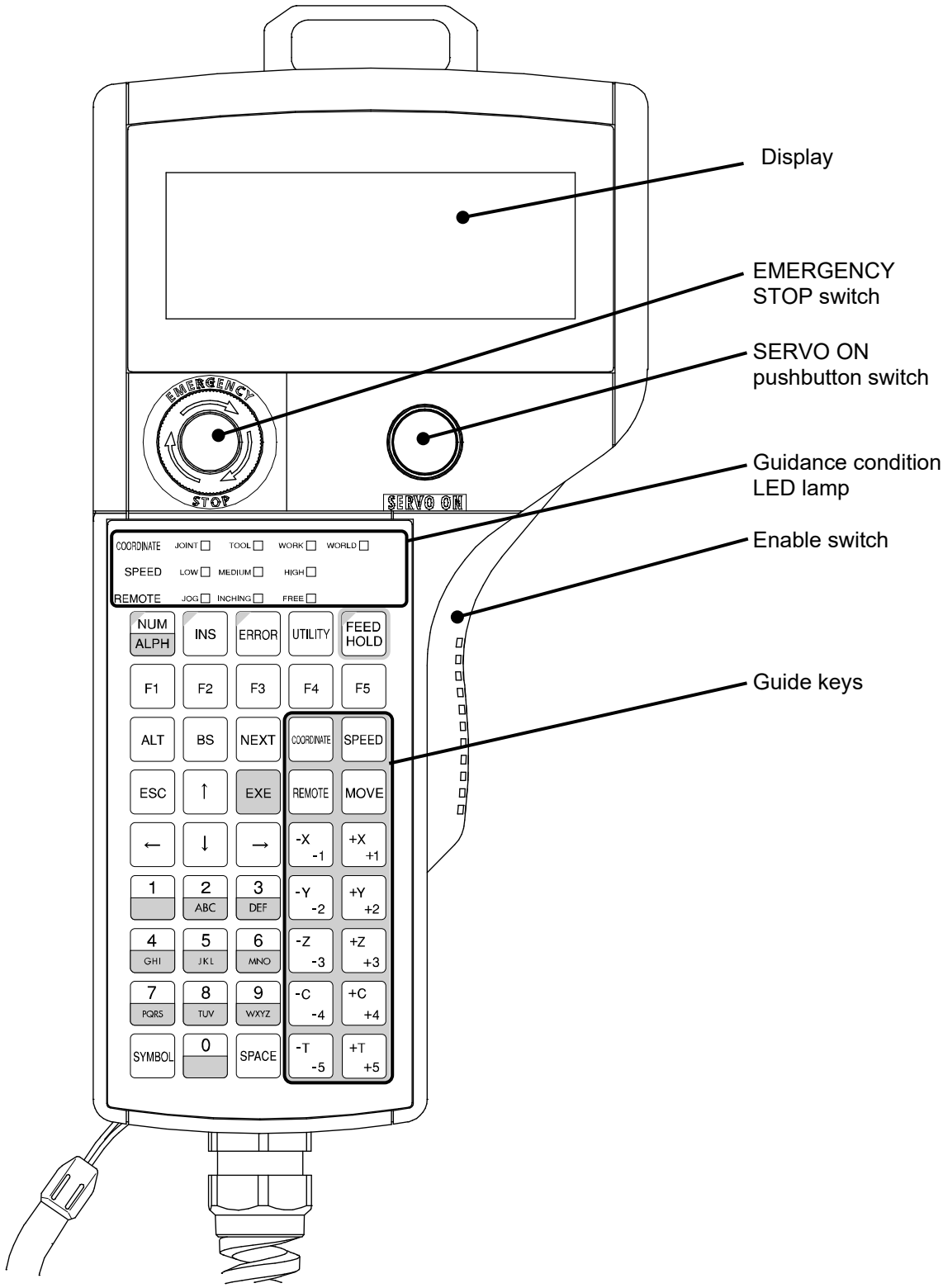
The control panel is shown below.




| | | |
|---|---|---|
| Key switch | : | Used to select the operation master mode. |
| MODE | : | Used to change over the EXT. SIGNAL and HOST modes while the EXT master mode is selected. |
| SERVO ON, SERVO OFF | : | Used to turn on and off the servo power. |
| RUN | : | Used to start the internal automatic mode operation. |
| STOP | : | Used to stop an operation after the active block of program has been executed. |
| CYCLE | : | Used to select the cycle operation mode in the internal automatic mode. |

| | | |
|---|---|---|
| SELECT | : | Each time this switch is pressed, the display content changes.<br>(LINE) No. of execution steps of program.<br>(USER) Value set in system variable PLCDATAW. (The value is input by the program.)<br>(ALARM) Alarm number generated in the controller. If two (2) or more alarms have generated, such alarm numbers are displayed at two (2)-second intervals. |
| UF1, UF2 | : | User function switches. This switch input can be referred to by the DIN command, and the lamp output by the DOUT command. |
| ALARM RESET | : | Used to reset all alarms generated. |
| EMERGENCY | : | EMERGENCY stop pushbutton switch. |

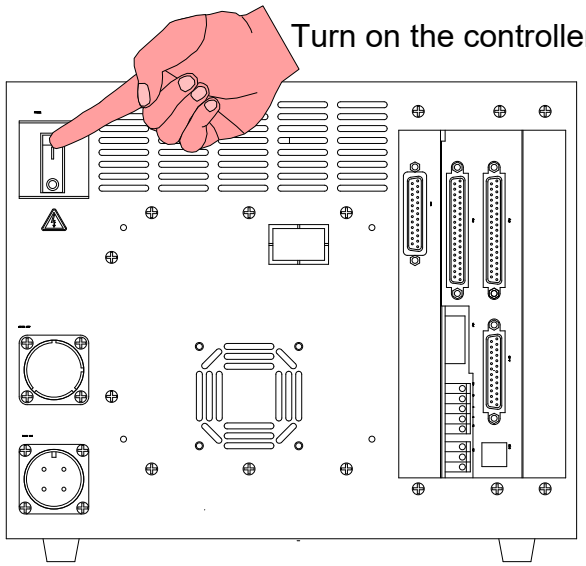
## 1.2 Teach Pendant



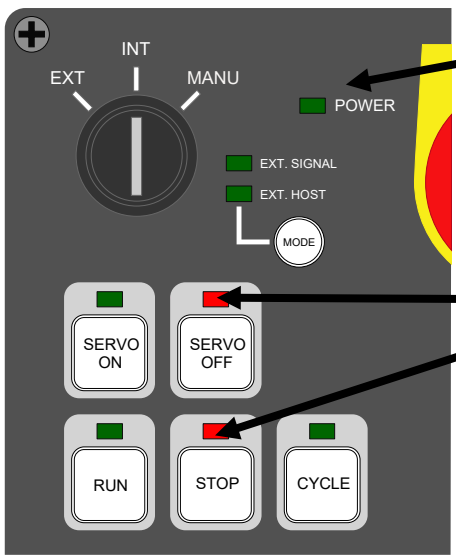
## 2. Power ON

### 2.1 Power ON

| | |
|---|---|
| ⚠ **CAUTION** | If an error has occurred or if the POWER LED equipped on the control panel will not illuminate when the main power switch of the controller was turned on, turn the power off immediately and check for the wiring. Otherwise, you will be involved in an electric shock or a fire.<br><br>When the power is turned on, the controller enters a program reset status. |



Turn on the controller power.




When the power is turned on, the POWER lamp is illuminated.

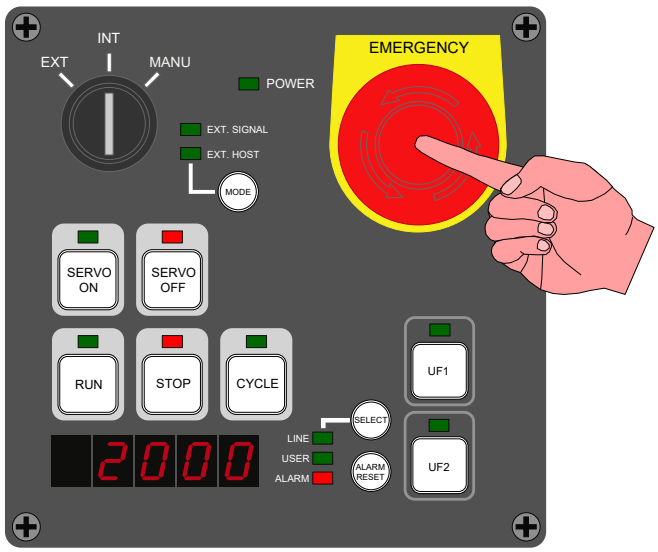
After power ON, the controller performs self-check. After the processing of start has finished, both the STOP and SERVO OFF lamps are illuminated in red.

## 2.2 Emergency Stop

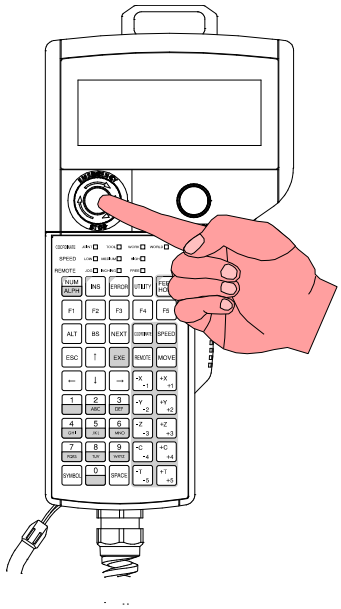
If a problem related to safety has occurred in the robot or peripheral equipment, press the EMERGENCY stop pushbutton switch immediately. The robot stops instantly with the servo power turned off.

[1] EMERGENCY stop switch

Control panel

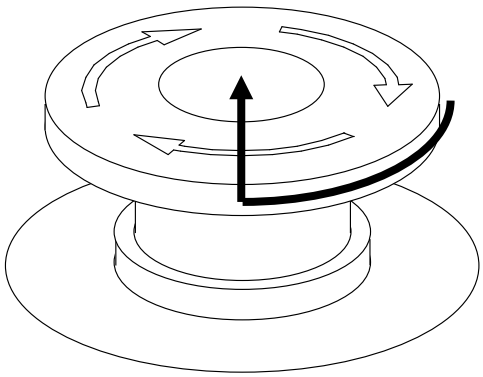



Teach pendant



- An EMERGENCY stop switch can be equipped on the peripheral equipment also. (For details, see the Interface Manual.)

[2] Reset of EMERGENCY stop switch



To reset the EMERGENCY stop switch, turn the top of the switch in the arrow-marked direction and pull it up.

## 2.3 Servo ON

| | |
|---|---|
| ⚠ **CAUTION** | If the robot has operated abnormally when the servo power was turned on, press the EMERGENCY stop switch immediately to turn the servo power off.<br><br>Before entering the working envelope of the robot, be sure to turn the servo power off. |

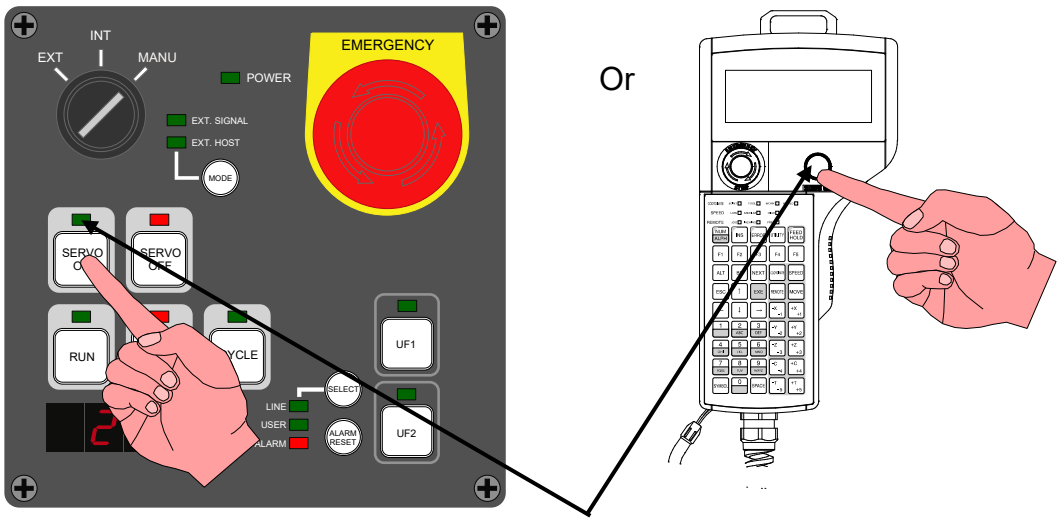
[1] Select "TEACHING" by means of the MODE selector switch.




Select "TEACHING" for the master mode by changing over the key switch.

[2] Make sure that all EMERGENCY stop switches (provided on the control panel, teach pendant and external equipment) are reset.

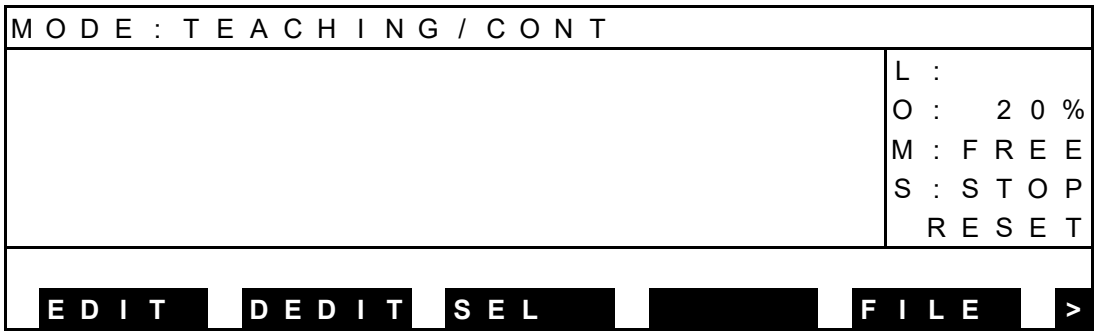
[3] Turn the servo power on.




Make sure that the SERVO ON lamp on the control panel or teach pendant is illuminated.

## 2.4 Main Display

Turn the power on, select the TEACHING mode by means of the MODE switch and turn the servo power on.




MODE: Indicates the master mode/operation mode.
The master mode comes in the following four (4) types.

| | |
|---|---|
| TEACHING | Teaching mode |
| INTERNAL | Internal automatic mode |
| EXT. SIGNAL | External automatic signal mode |
| INT. HOST | External automatic host mode |

For the operation mode, the following four (4) modes are available.

| | |
|---|---|
| CONT | Continuous operation |
| CYCLE | Cycle operation<br>(The program is executed from the top to the end command of the main program.) |
| SEG | Segment operation<br>(The program is executed just before the next operation command.) |
| STEP | Step operation<br>(The program is executed line by line.) |

L: (Line)
Indicates the active program line number.

O: (Override)
Indicates the speed override value, taking the specified speed as 100 %.

M: (Machine)
On the TS1000 screen, "FREE" is always indicated.

S: (Status)
Indicates the program run status of the controller.
The run status comes in the following four (4) types.

| | |
|---|---|
| RUN | During program execution |
| STOP RESET | During stop of initialization mode (When the RUN mode takes effect, the program is executed from the top.) |
| STOP RETRY | During stop of restart mode (When the RUN mode takes effect, interrupted motion command restarts.) |
| STOP CONT | During stop of continuous mode (When the RUN mode takes effect, the program is executed from the command next to the finished command.) |

Function: The function menu is displayed by reversed image on the bottom of the display. The keys correspond to F1 to F5 in turn, when viewed from the left. If symbol ">" is displayed on the bottom right side of the display, the next function menu can be displayed by means of the NEXT key.

## 2.5 Power OFF

[1] Make sure that the robot is not working.




Make sure that the red-colored STOP lamp equipped on the control panel is illuminated.

Unless the STOP lamp is illuminated, press the STOP switch and wait until the robot motion stops with the STOP lamp illuminated.

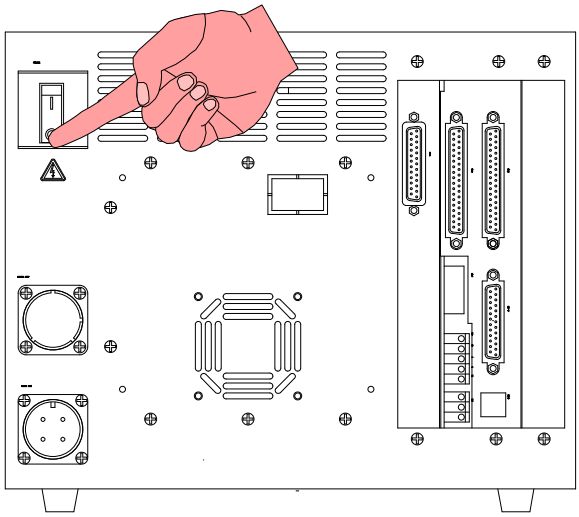
[2] Turn the servo power off.




Press the SERVO OFF switch to turn the servo power off.

Make sure that the SERVO OFF lamp is illuminated in red.

[3] Turn off the controller power.

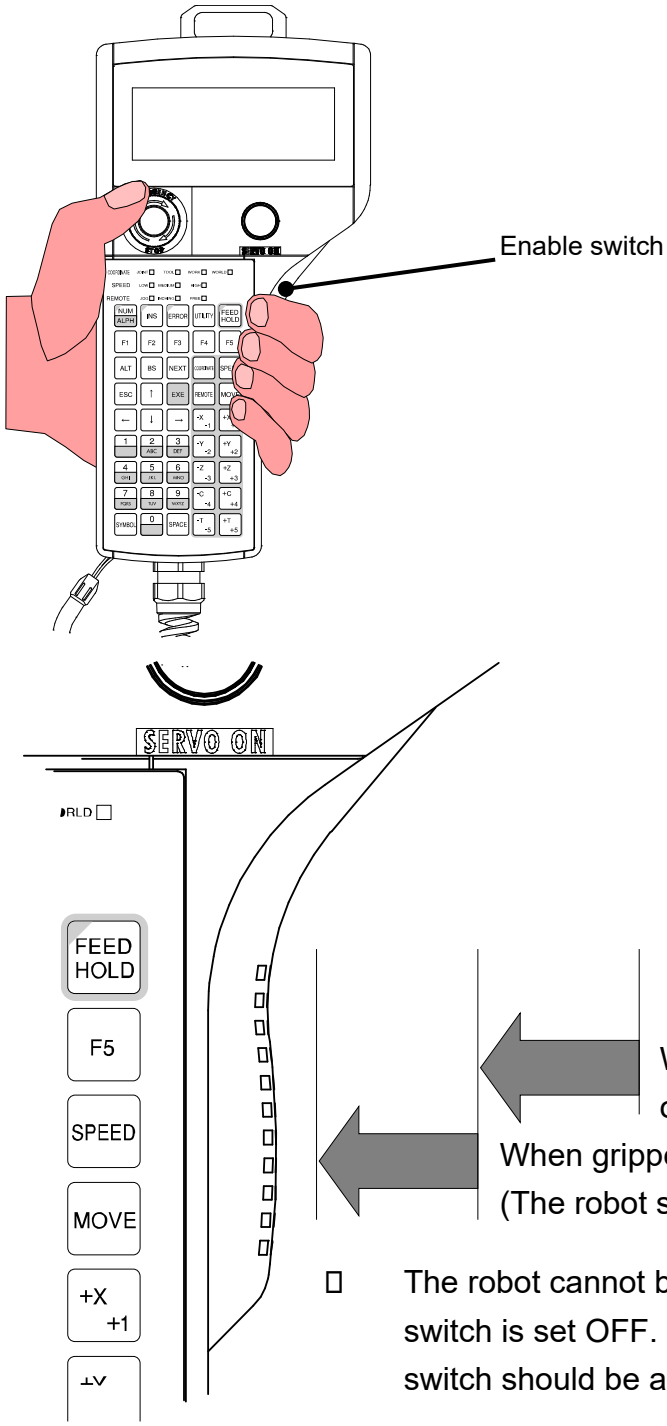


Turn off the controller power.

## 3. Manual Operation

### 3.1 Enable Switch

To manually guide the robot, set ON the enable switch.




When gripped light, the switch turns on.

When gripped strong, the switch turns off. (The robot stops)

- The robot cannot be guided manually if the enable switch is set OFF. To manually guide the robot, this switch should be always set ON.

### 3.2 Joint Guide

[1] Turn on the controller power, select "TEACHING for the master mode and turn on the servo power.

[2] Select the guidance coordinate system.
Press the COORDINATE key to select the guidance coordinate system.
Each time the key is pressed, the status of the guidance coordinate system select lamps on the keyboard changes as shown below.

→ JOINT→ TOOL→ WORK → WORLD ┐

Select JOINT□ now.

[3] Select the guide speed.
Press the SPEED key to select the guide speed.
Each time the key is pressed, the status of the guide speed select lamps on the keyboard changes as shown below.

→LOW →MEDIUM →HIGH ┐

Select LOW□ now.

[4] Select the guidance mode.
Press the REMOTE key to select the guidance mode.
Each time the key is pressed, the status of the guidance mode select lamps on the keyboard changes as shown below.

→JOGn →INCHINGo → FREEo ┐

Select JOG□ now.

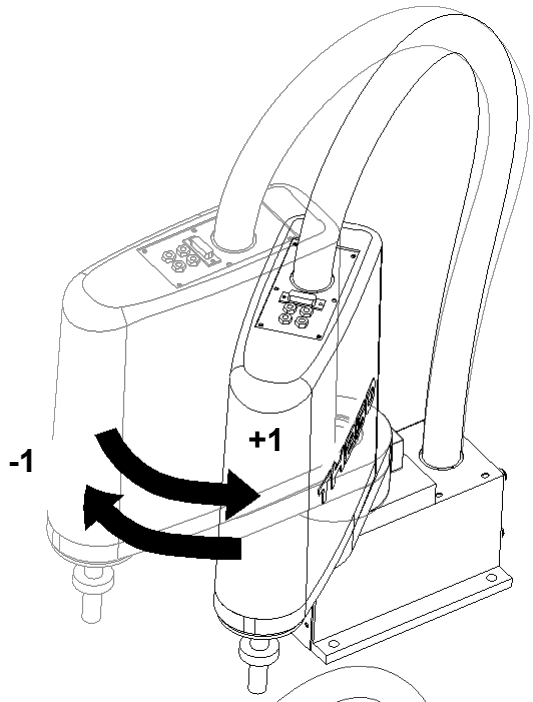
[5] Set ON the enable switch.

[6] Press the +X +1 key, and the axis 1 moves in the plus (+) direction.

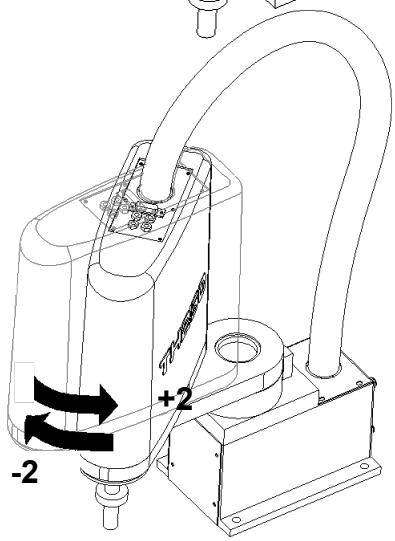
[7] Press the –X –1 key, and the axis 1 moves in the minus (–) direction.

[8] Press the SPEED key to change the guide speed, then move the robot in the plus (+) and minus (–) directions by means of the [+X +1] and [–X –1] keys.

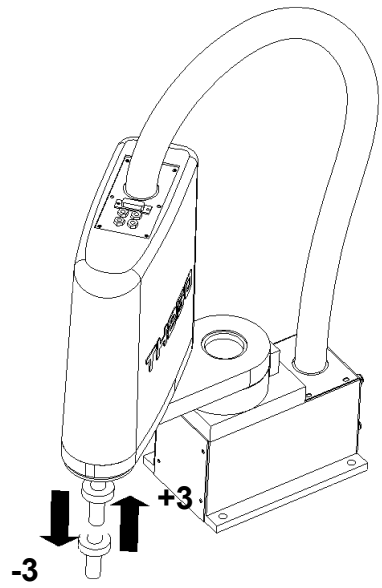
[9] Press the REMOTE key to change the guidance mode to "INCHING", and move the robot by inching, using the [+X +1] and [–X –1] keys.



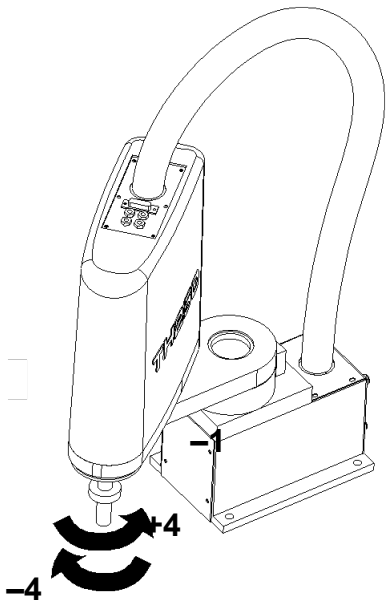

[10] Move the axis 2 by means of the [+Y +2] and [–Y –2] keys.




[11] Move the axis 3 by means of the [+Z +3] and [–Z –3] keys.



[12] Move the axis 4 by means of the +C +4 and –C –4 keys.



### 3.3 Guide in Cartesian Coordinate System

[1] Select the guidance conditions.
Press the COORDINATE key to select "WORLD".
Press the SPEED key to select a guide speed.
Press the REMOTE key to select "JOG" or "INCHING".

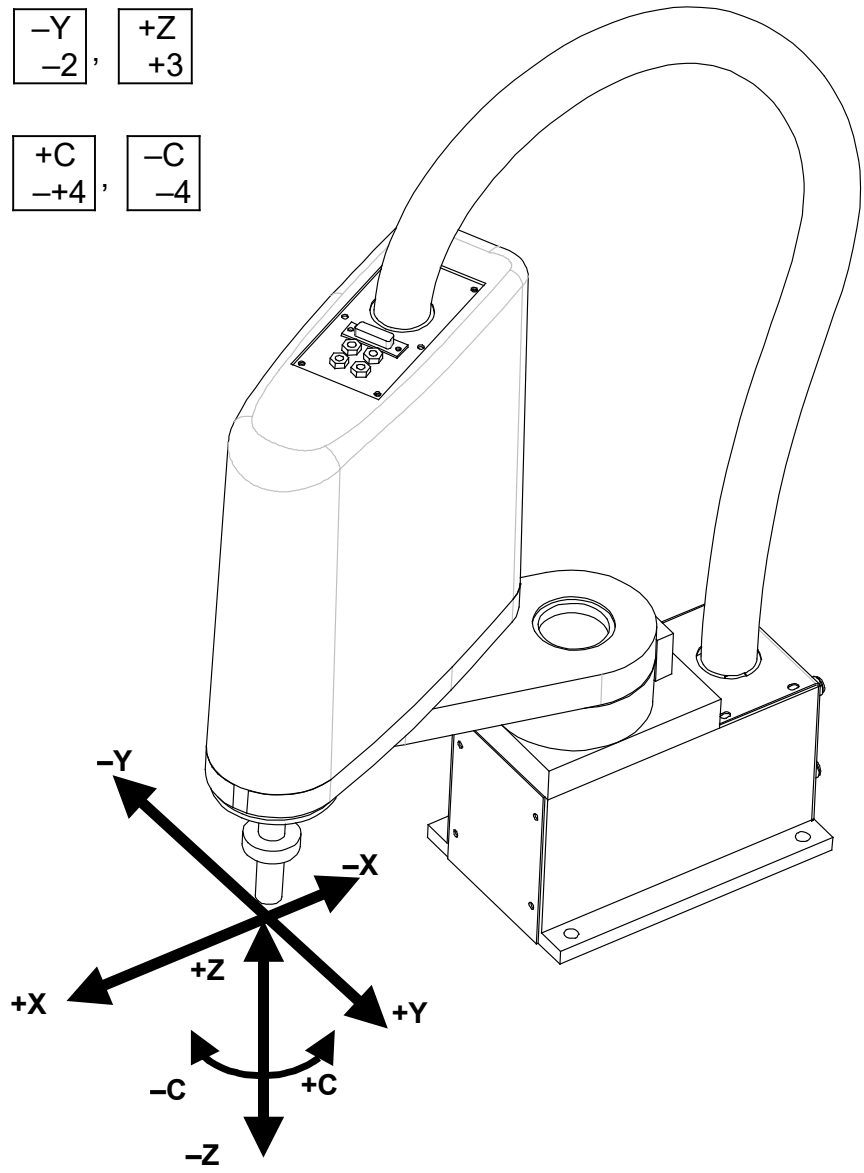
[2] Set ON the enable switch.

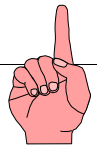
[3] Move the robot in the X-axis direction by means of the [+X +1] and [–X –1] keys.

[4] Move the robot in another axis directions by means of the following keys.

[+Y +2], [–Y –2], [+Z +3]

[–Z –3], [+C –+4], [–C –4]





**One point**

## Display of Current Position during Manual Guidance

[1] Press the UTILITY key, and the teach pendant display as shown below is called.

```
ROBOT  UTILITY


AUX   I/O   POS   TRANS   WK-TM   >
```

[1] Press the [F3] key. (Selection of [POS])

```
JOINT
J1:        0.000
J2:        0.000
J3:        0.000
J4:        0.000
J5:        0.000

JOINT  WORLD  WORK  P-CMD  P-FBK
```

Positions (i.e., angles) of respective joints are displayed.

[3] Press the [F2] key. (Selection of [WORLD])

```
WORLD
  X:    250.000      WORK:.world
  Y:      0.000      TOOL:
  Z:      0.000      BASE:.world
  C:      0.000
  T:      0.000

JOINT  WORLD  WORK  P-CMD  P-FBK
```

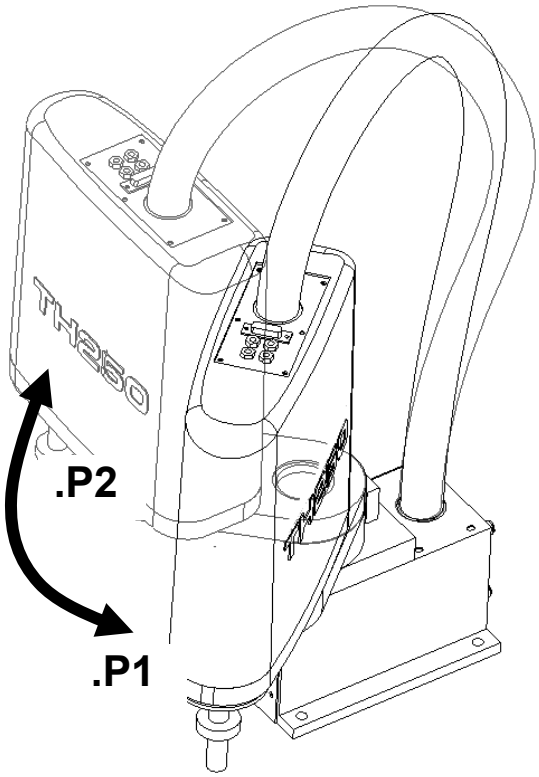
The current position in the world coordinate system (i.e., Cartesian coordinate system) is displayed.

[4] To display the joint coordinates again, press the [F1] key. (Selection of [JOINT])

[5] To quit, press the ESC key twice.

## 4. Programming

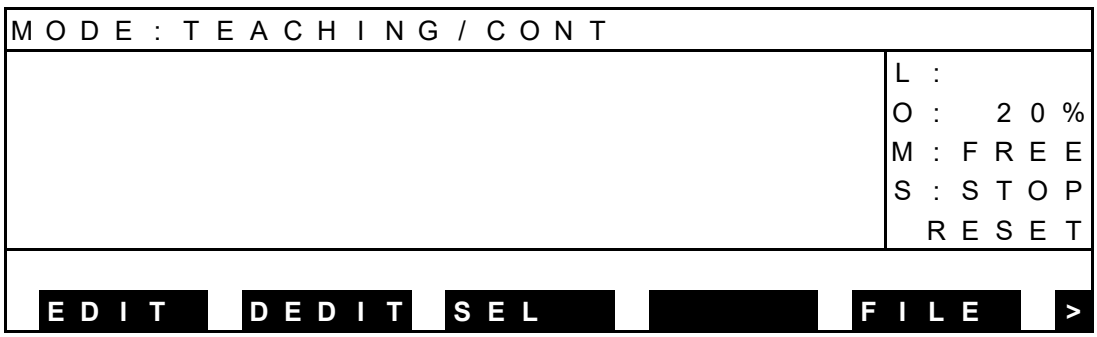
Now, let's create a simple program in which the robot moves from P1 to P2.




### 4.1 Starting Editor

[1] Press the MODE key equipped on the control panel to select "TEACHING" for the master mode.

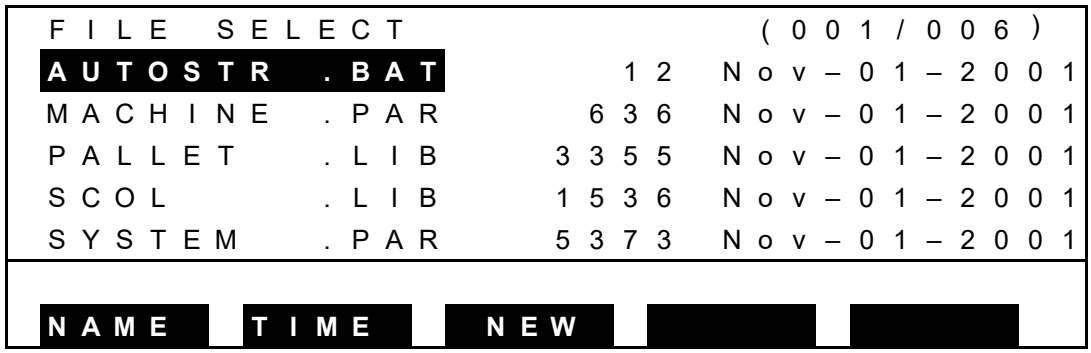
[2] Make sure that [EDIT] is assigned to the [F1] key of the function menu.



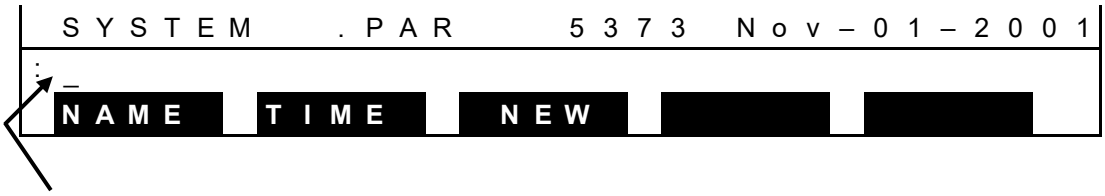

Unless [EDIT] is displayed, press the NEXT key a few times until [EDIT] is displayed.

[3] Press the [F1] key and start the editor.

[4] When [EDIT] is selected, the file select screen appears.




[5] Press the [F3] key to create a new file. (Selection of [NEW])




The cursor is displayed.
Enter the file name of a new program you wish to create.
(The file name can be specified by up to (8 + 3 characters; ???????.???) alphanumeric characters.)

[6] Press the [NUM ALPH] key to select the ALPH mode. (The switch lamp goes out.)

[7] To enter file name "TEST", for instance, observe the following steps.

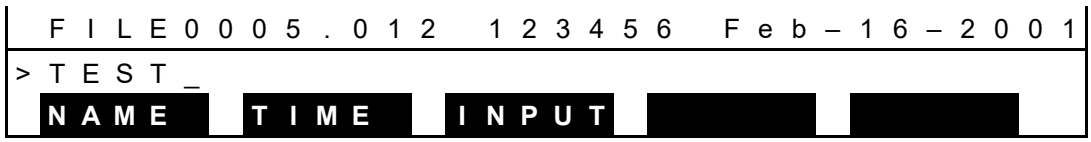
Press the keys [8 TUV], [8 TUV], [EXE] to enter alphabet "T".

Press the keys [3 DEF], [3 DEF], [3 DEF], [EXE] to enter alphabet "E".

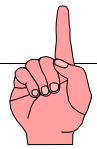
Press the keys [7 PQRS], [7 PQRS], [7 PQRS], [7 PQRS], [7 PQRS], [EXE] to enter alphabet "S".

Press the keys [8 TUV], [8 TUV], [EXE] to enter alphabet "T".

The display changes as shown below.




[8] Press the EXE key, and new program "TEST" is selected and the program editor starts.



## Input of Numerals and Alphabets

**One point**

The alphabetic input and numeric input modes can be changed over alternately by pressing the [NUM ALPH] key.

When the numeric input mode is selected, the key lamp is turned on.

[1] Numeric key input
Select the numeric input (NUM) mode and press an appropriate numeric key.

Example: To input number "8", press [8 TUV] key. Then number "8" is input.

[2] Input of alphabet/symbol key
Select the alphabetic input (ALPH) mode. As alphabets are grouped, press the relevant key until a desired alphabet is displayed. After the alphabet is displayed, press the EXE key, right cursor key or any key of another group, and it is established.
In the ALPH mode, a number is also displayed firstly, and it is possible to input it in the ALPH mode.

Example 1: To enter alphabet "B", press the [2 ABC] key three (3) times. When "B" is displayed, press the EXE key.
Unless "B" is determined, it is displayed by reversed image. Once established, it is indicated normally.

Example 2: To enter a symbol, press the SYNBOL key. Each time this key is pressed, a choice of symbol is displayed. When the desired symbol is displayed, press the EXE key.

- Though symbols can be input both in the NUM and ALPH modes, choices differ with the mode selected then.

[3] Keys effective during unsettlement

[EXE] [→] ····· Establishment of an undecided character.

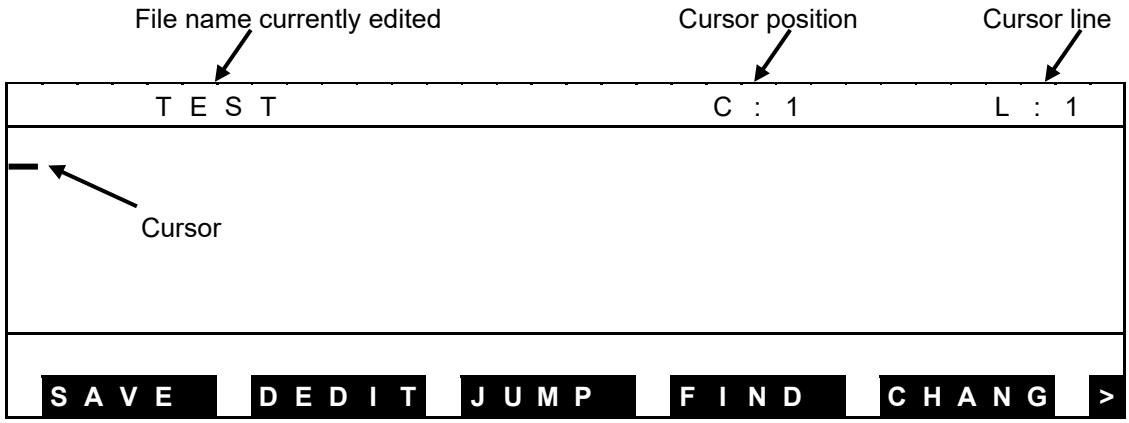
[ESC] [BS] [←] ····· Cancel of an undecided character.

[↑] ····· Previous choice of undecided character.

[↓] ····· Next choice of undecided character.

## 4.2 Entering Program

[1] Program editor

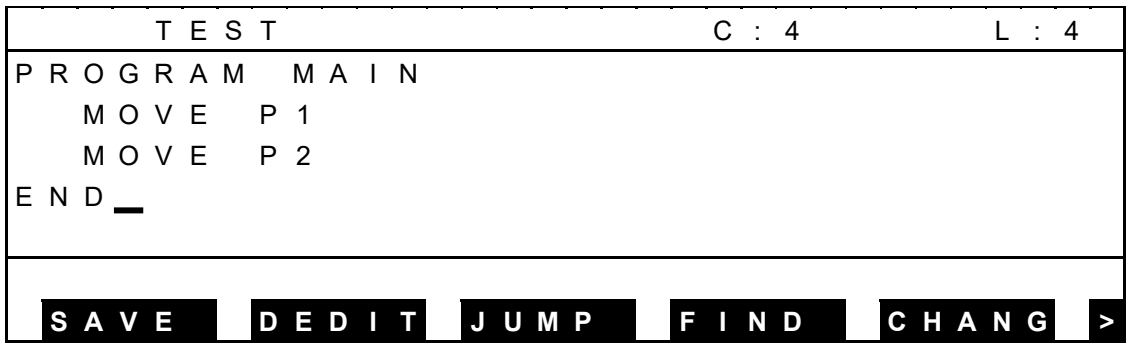



[2] Input of program

Let's enter the following program consisting of four (4) lines (or blocks).

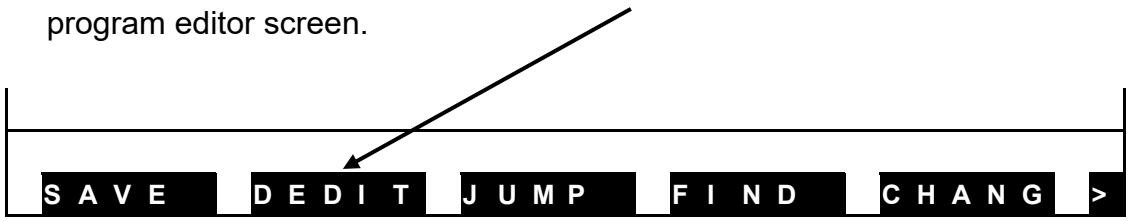
| | | |
|---|---|---|
| PROGRAM MAIN | ····· | Start of main program |
| MOVE P1 | ····· | Move to point P1 |
| MOVE P2 | ····· | Move to point P2 |
| END | ····· | End of main program |

[3] When the program is entered, the following screen appears.



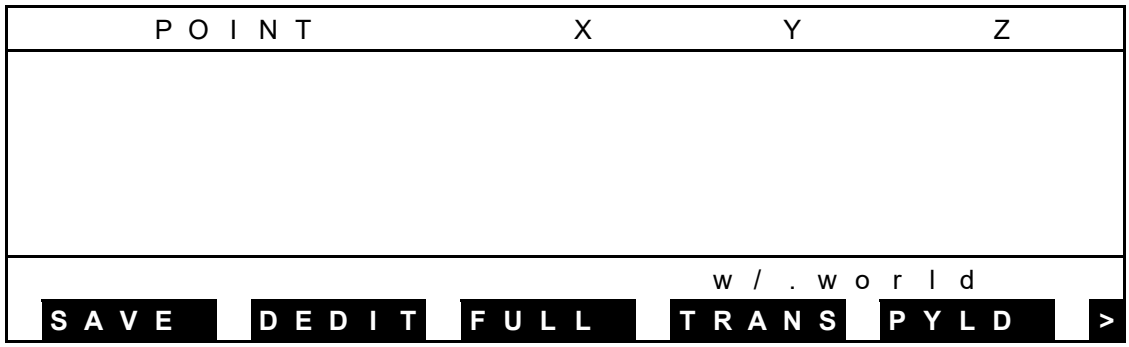
### 4.3 Entering Data (Teach Points)

[1] Start of data editor
Make sure that "DEDIT" is assigned to the [F2] key of the function menu on the program editor screen.



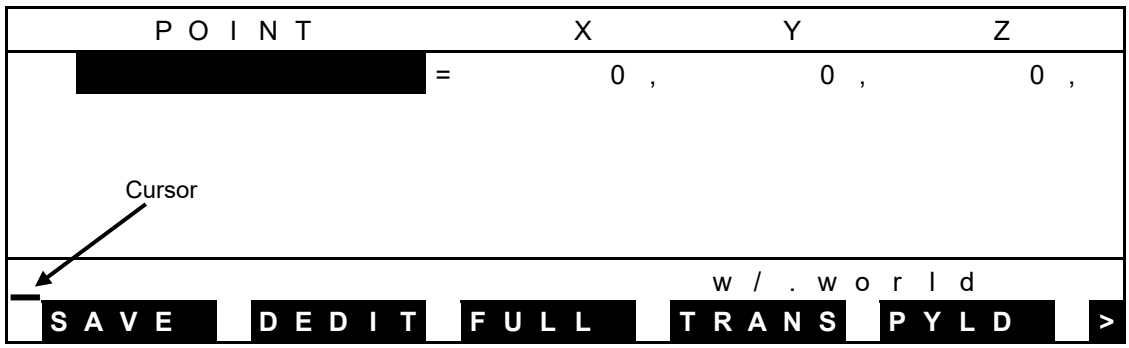

Unless [DEDIT] is displayed, press the NEXT key a few times until [DEDIT] is displayed.

[2] Press the [F2] key and start the editor.




- As the data to be displayed is absent, only the title is displayed.

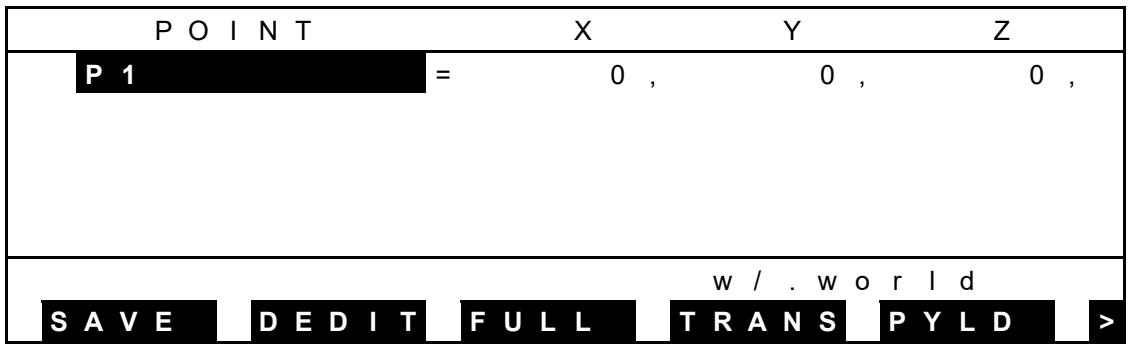
[3] To add a new teach point data, press the INS key. The following screen is called with the cursor displayed.



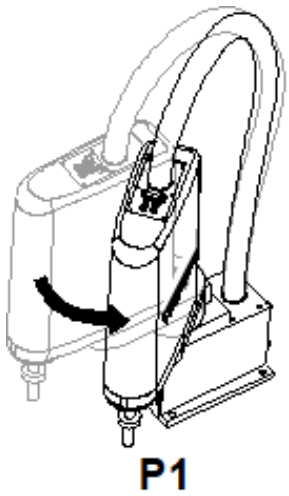
[4] Enter teach point name "P1" and press the EXE key.



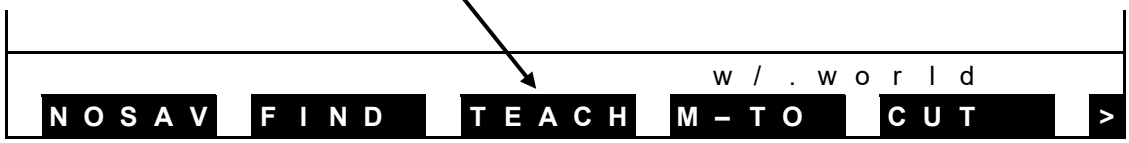

[5] Teach point name "P1" has been registered.




[6] Manually guide the robot to the position you wish to teach.




[7] Press the NEXT key to display [TEACH].

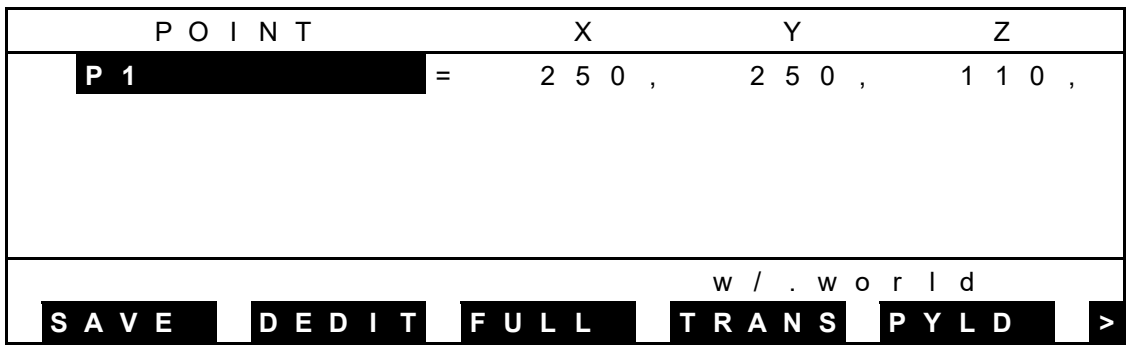


[8] Press the [F3] key and teach the current robot position.

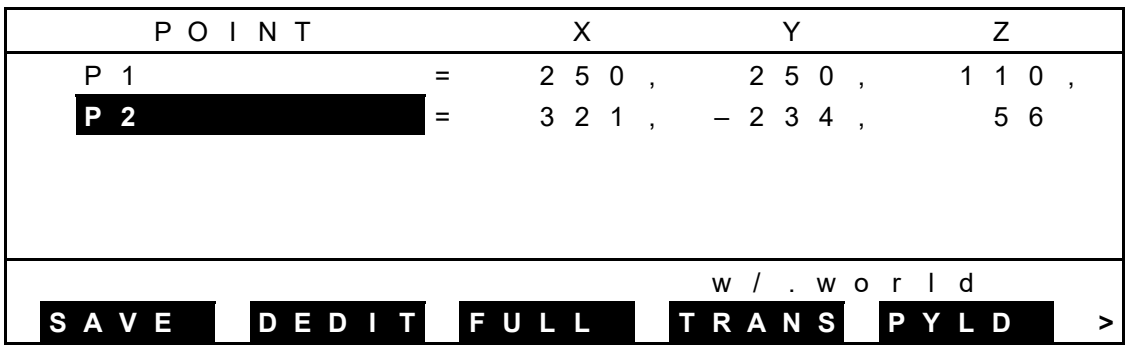


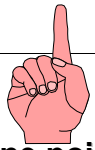

When the above screen appears, press the EXE key. (To cancel, press another key.)

[9] Now, the current robot position has been taught.




[10] Press the INS key and add the "P2" data.
The procedures are the same as in the "P1" data. Observe Steps [3] through [9] above. (Manually guide the robot to a position which is different from "P1".)





**One point**

## Detailed display of teach points

To display the teach points in detail, press the [FULL] key of the function menu.

```
P 1                                          5 /    1 0 ]
X:     3 0 0 . 0 0 0  C O N F I G : F R E E
Y:     3 0 0 . 0 0 0
Z:     3 0 0 . 0 0 0
C:     3 0 0 . 0 0 0
T:     3 0 0 . 0 0 0
P O I N T 0 0 0 0 5                  w / . w o r l d
 SAVE   PRG    DIR    TRANS   PYLD    >
```

To return the system to the directory display, press the [DIR] key of the function menu.

## 4.4 Quitting Editor

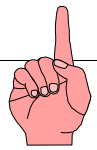
[1] Press the NEXT key a few times to display [SAVE].




[2] Press the [F1] key to select [SAVE].




When the above screen is displayed, press the EXE key. (To cancel, press another key.)



**One point**

### Abortion of file during editing

Press the NEXT key a few times to display [NOSAV].




Press the [F1] key to select [NOSAV].




When the above screen is displayed, press the EXE key. (To cancel, press another key.)

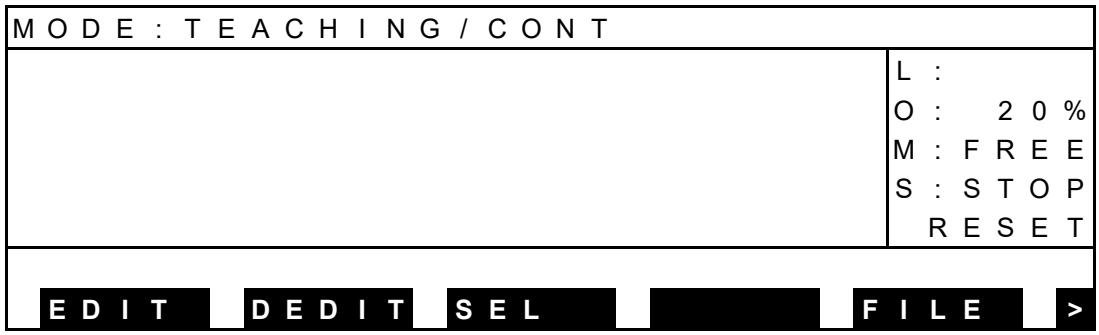
## 5. Starting Program

### 5.1 Selecting Program

Select program "TEST" which was created in Section 4 above.

[1] Select "TEACHING" for the master mode by pressing the MODE switch provided on the control panel.

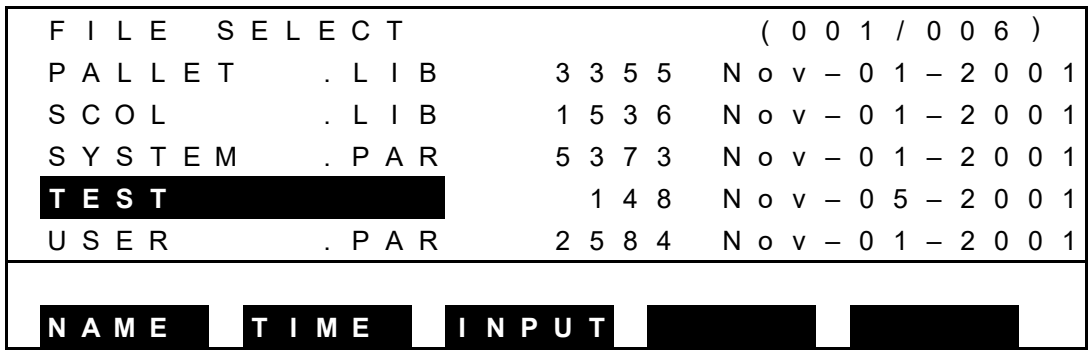
[2] Make sure that [SEL] is assigned to the [F3] key of the function menu.



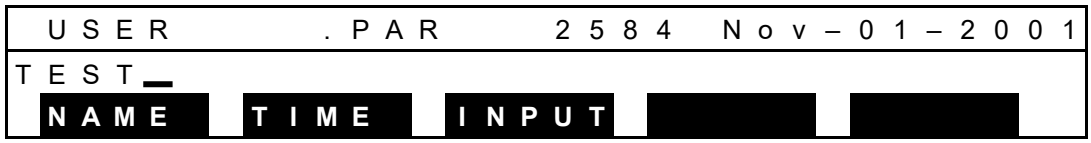

Unless [SEL] is displayed, press the NEXT key a few times until [SEL] is displayed.

[3] Press the [F3] key to open the file select screen.

[4] Inversely display "TEST", using the ↓ and ↑ keys.

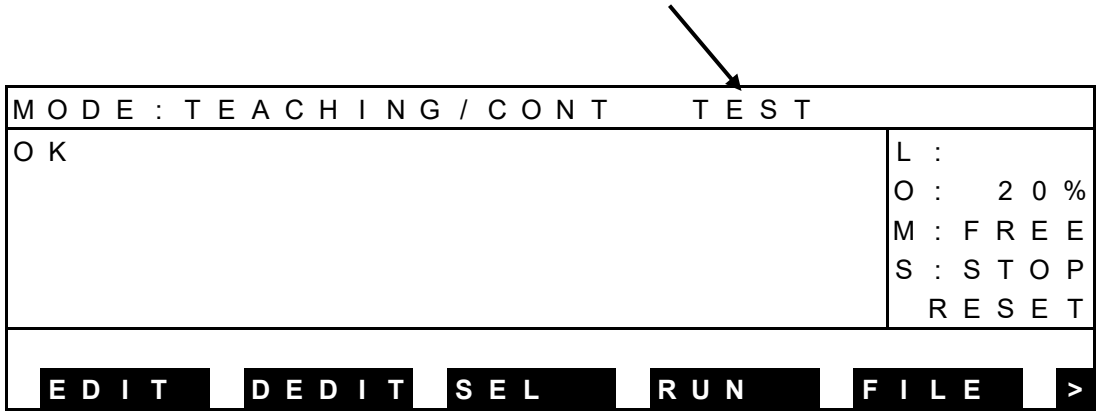


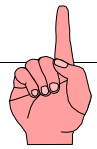
[5] Press the EXE key to select program "TEST".




[6] When "TEST" is displayed, press the EXE key again.

[7] When the program has been selected correctly, the file name of the selected program is displayed at the top of the screen.





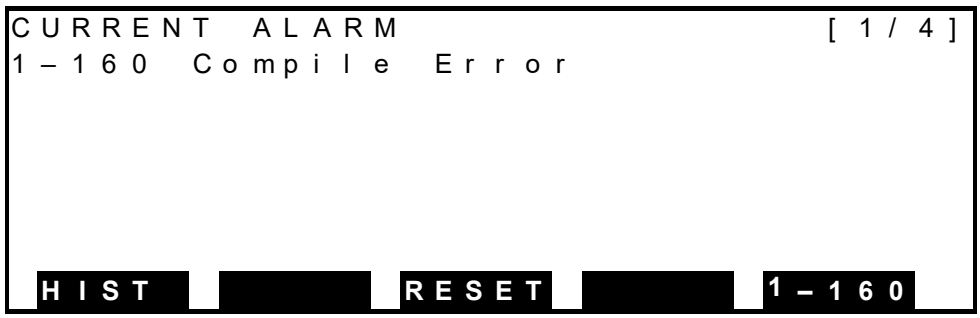
**One point**

## Program cannot be selected.

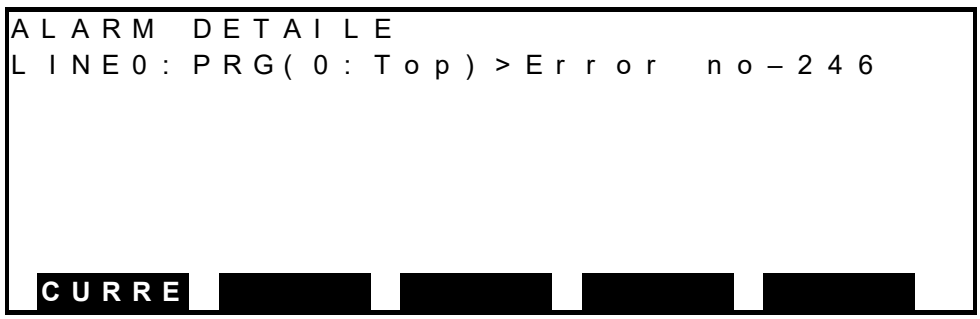
Once the program has been selected, it is compiled.
("Compile" signifies the program check and conversion of the program into a format which the robot controller can operate arithmetically at high speed.)

If an error is found in the program at this time, an alarm is generated without selecting the program.

Press the ERROR key on the teach pendant, and the alarm currently generated is displayed.



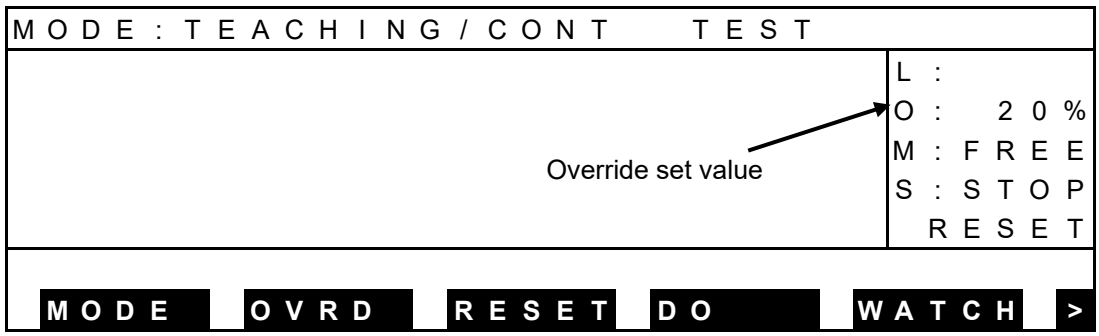

To see the details of the error, press the [F5] key.



## 5.2 Test Operation

A programmed operation in the TEACHING mode is called the "test operation."

[1] Change the speed override.
The current override value is shown by "O: %" on the right side of the screen.



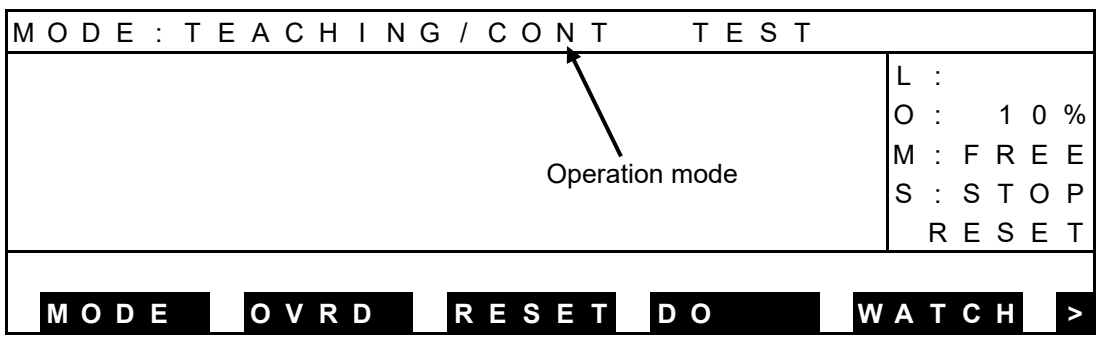

Press the NEXT key to assign [OVRD] to the [F2] key of the function menu.
Press the [F2] key then.




Select the 10 % override by pressing the [F2] key followed by the EXE key.

For the test operation, the speed override should be less than 25 %, which can be changed by appropriate parameter.
In this step, the override value was set, using the function key. It is also possible, however, to directly input the value.

[2] Select "STEP" for the operation mode.

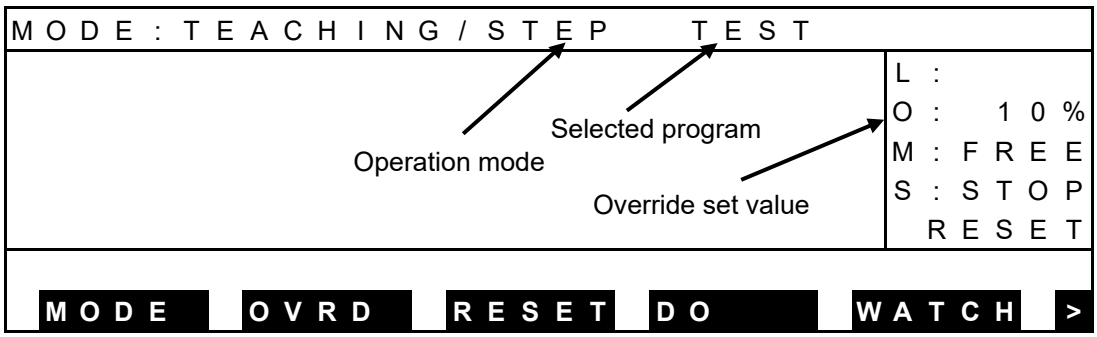


Press the NEXT key to assign [STEP] to the [F4] key of the function menu. Press the [F4] key then.




Press the EXE key to select the step operation.

[3] Start the step operation.
Before starting the operation, confirm the selected program name, operation mode and override set value.



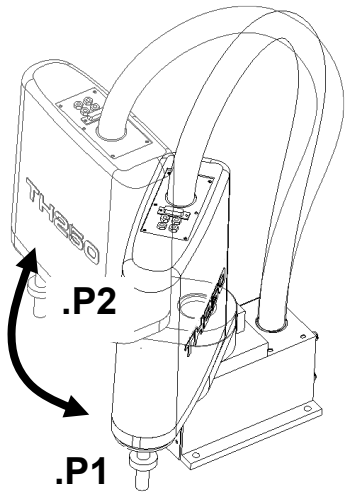

Press the NEXT key to assign [RUN] to the [F4] key of the function menu. Press the [F4] key then.




Press the EXE key, and the program to be executed is displayed.




Press the EXE key, and the displayed program step is executed with the subsequent step displayed.




```
                                                    RESET
MOVE   P2
 EDIT   DEDIT   SEL    RUN    FILE   >
```

To terminate the step operation, press the EXE key.

[4] Cycle operation

Press the NEXT key to assign [CYCLE] to the [F2] key of the function menu.
Press the [F2] key then.

```
                                                    RESET
MODE : CYCLE
 CONT   CYCLE   SEG    STEP          >
```

Press the EXE key to select the cycle operation.
Before starting the operation, confirm the selected program name, operation mode and override set value.
Press the NEXT key to assign [RUN] to the [F4] key of the function menu.
Press the [F4] key then.

```
                                                    RESET
RUN
 EDIT   DEDIT   SEL    RUN    FILE   >
```

Press the EXE key, and the cycle operation starts.

The robot executes to the end command of the program and stops.

[5] Press the NEXT key to assign [CONT] to the [F1] key of the function menu.
Press the [F1] key then.




Press the EXE key to select the continuous operation.
Before starting the operation, confirm the selected program name, operation mode and override set value.
Press the NEXT key to assign [RUN] to the [F4] key of the function menu.
Press the [F4] key then.




Press the EXE key, and the continuous operation starts.

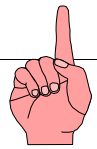
The robot repeats the program from the top to the end command.

[6] Stop of program execution.
Press the NEXT key to assign [STOP] to the [F4] key of the function menu.
Press the [F4] key then.




The robot stops executing the program after the current command has been executed.

You can stop the program also by pressing the STOP key provided on the control panel.
To execute the program again, select [RUN] from the function menu.



**One point**

## BREAK and FEED HOLD

To stop the ongoing robot operation, the break and feed hold functions are also available, in addition to the stop function.

To use the break function, select [BREAK] from the function functions. Then the robot will decelerate and stop even during the motion command.
To resume the operation, select [RUN] from the function menu.

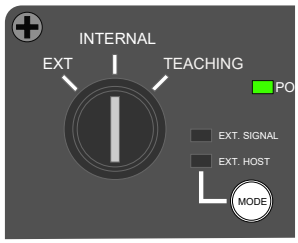
To select the feed hold function, press the FEED,HOLD key equipped on the teach pendant. (The key lamp is illuminated then.)
The robot will slow down and stop even during the motion command.
To restart the operation, press the FEED,HOLD key again.

Caution at restart of operation:

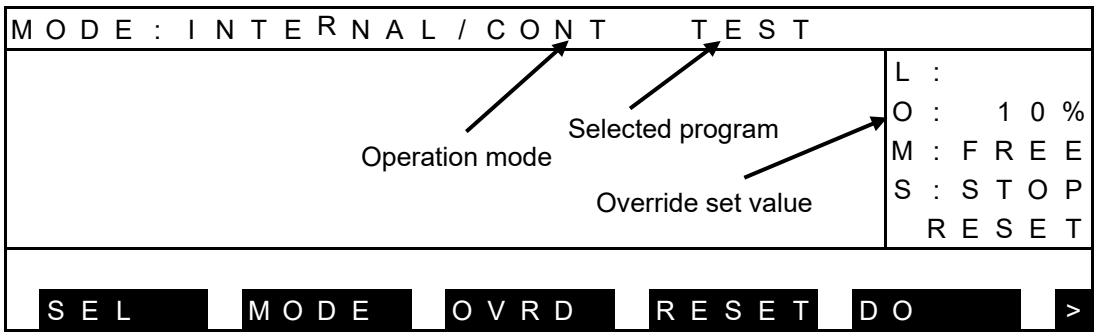
At the time of restart, the robot restarts the operation toward the target position of the interrupted motion in both the break and feed hold modes. Its path may differ from the normal operation, however.

## 5.3 Internal Automatic Operation

[1] Change over the key switch provided on the control panel to select "INTERNAL" for the master mode.




[2] Before starting the operation, confirm the selected program name, operation mode and override set value.

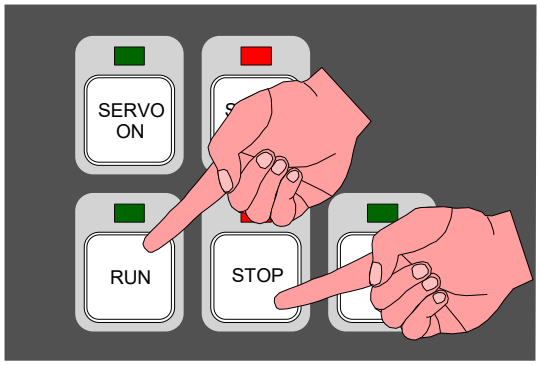



[3] Confirm the selected program. The file selected in the TEACHING mode is taken over. (Unless selected, select a desired file, using [SEL] of the function menu.)

[4] Confirm the override value. In the TEACHING mode, the value entered in the TEACHING mode remains effective. (Unless entered in the TEACHING mode, the override is 100 %. Set the override, using [OVRD] of the function menu.)

[5] Select the execution mode by means of the [MODE] key of the function menu.

[6] In the internal automatic mode, program start and stop are performed through the control panel.

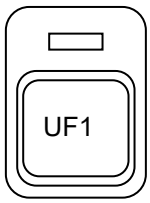



[7] Change the override value, using [OVRD] of the function menu.
In the INTERNAL mode, the override of up to 100 % can be entered.
Unless the speed will reach 100 %, perform check on system signal LOW_SPD (low-speed command). For details, see the Interface Manual.

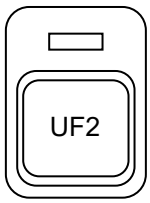
## 6. Input/Output Control

You can use the UF1 and UF2 switches and LEDs equipped on the control panel as the user's I/Os.
This section describes the programs, using the I/Os of the control panel.
For the other I/Os (i.e., standard input/output and hand input/output), see the Interface Manual.

### 6.1 User's Function Switches

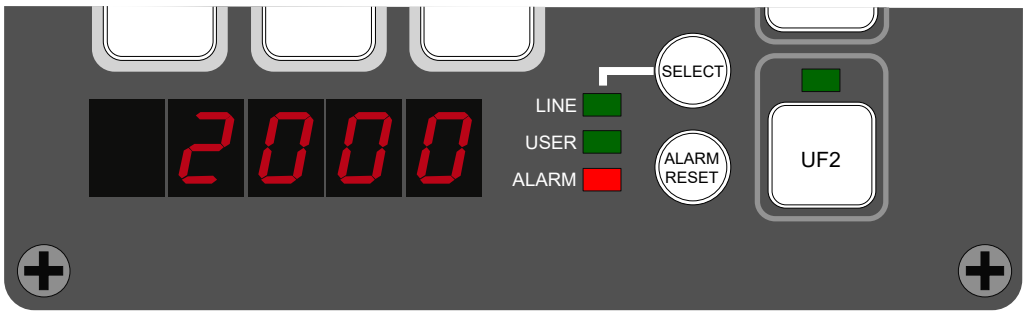



To be output by the DOUT (63) command.
To be referred to by the DIN (63) command.




To be output by the DOUT (64) command.
To be referred to by the DIN (64) command.

### 6.2 7-Segment User Display




When "USER" is selected by means of the SELECT switch, value of system variable PLCDATAW1 is displayed.

If "PLCDATAW1 = 10" is programmed, "10" is displayed.

- The range of displayed value is 0 ~ 65535. If a value exceeding this range is specified, it will not be displayed correctly.

## 6.3 I/O Control Program

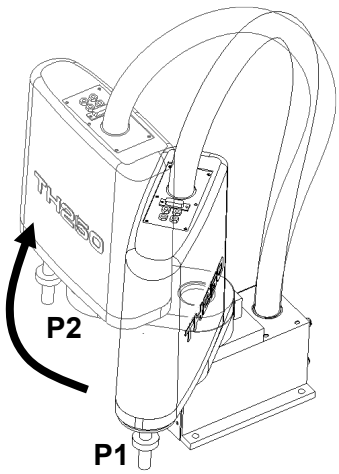
[1] Program using an input signal

Now, let's change the program created in Section 4.

```
PROGRAM MAIN        ·····  Start of main program
    MOVE P1         ·····  Move to point P1
    MOVE P2         ·····  Move to point P2
    WAIT DIN (63)   ·····  Wait until input 63 (UF1 switch) turns on.
END                 ·····  End of main program
```

Execute the test operation. Select "CONT" (continuous operation) for the operation mode.

After the start of the program, the robot stops at point P2.




Press the UF1 switch, and the robot moves to point P1.

After moving to point P2 again, the robot stops and waits for the input of the switch.

[2] Program using an output signal
Now, let's change the program in Para. [1] above.

```
PROGRAM MAIN          ·····  Start of main program
    MOVE P1           ·····  Move to point P1
    DOUT (63)         ·····  Output 63 (UF1 lamp) turns on.
    MOVE P2           ·····  Move to point P2
    WAIT DIN (63)     ·····  Wait until input 63 (UF1 switch) turns on.
    DOUT (–63)        ·····  Output 63 (UF1 lamp) turns off.
END                   ·····  End of main program
```

Execute the test operation. Select "CONT" (continuous operation) for the operation mode.
After the start of the program, the UF1 lamp is illuminated at the same time that the robot has reached point P1. After moving to point P2, the robot stops and waits for the input of the switch. Press the UF1 switch then. The UF1 lamp goes out and the robot continues the operation.

[3] Output to 7-segment user display
Now, let's change the program as created in Section 4.

```
PROGRAM MAIN          ·····  Start of main program
    PLCDATAW1 = 1000  ·····  A value is written to the 7-segment user
                             display.
    MOVE P1           ·····  Move to point P1
END                   ·····  End of main program
```

Execute the test operation. Select "CONT" (continuous operation) for the operation mode and start the program.

Select "USER" by means of the SELECT switch equipped on the control panel, and the value of "1000" is displayed on the 7-segment user display.